Results

Table 1 and Figure 1 compare the three methods on the NYC Taxi Fare dataset (55 423 856 rows). Chunking and Dask gave the same mean fare, 11.35.

Method	Time (s)	Peak RAM (MB)
Standard pandas	291.8	9 117
Pandas chunking	60.2	159
Dask DataFrame	26.1	733

Table 1: Execution time and peak RAM of the three methods.

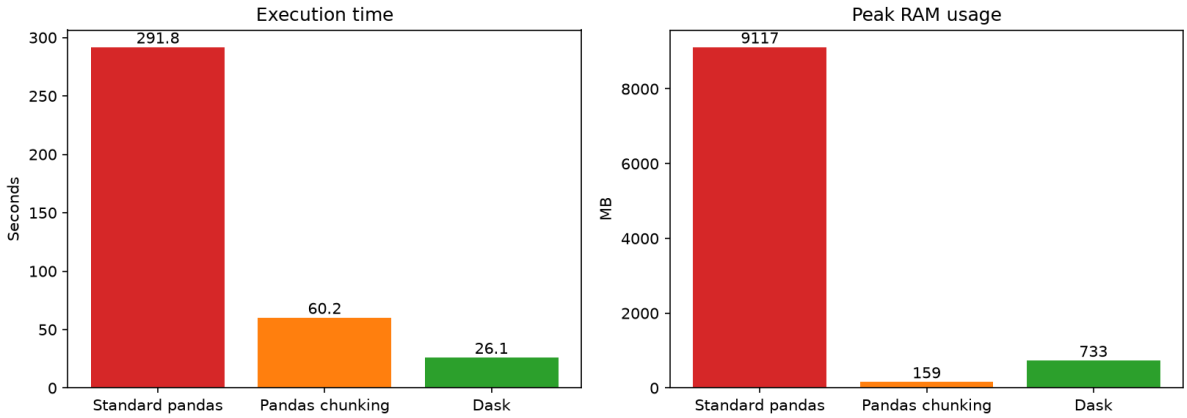


Figure 1: Execution time (left) and peak RAM (right) of the three methods.

Standard pandas was the slowest and used about 9.1 GB of RAM. It did not raise a `MemoryError`, because the machine has enough memory(12GB of ram). Pandas chunking kept memory low (159 MB) but processed the chunks one after another. Dask was the fastest, about 2.3 times faster than chunking, and used more RAM than chunking (733 MB) because several partitions are in memory at once.